

OTTAWA TRANSIT ACCESSIBILITY ANALYSIS

METHODOLOGY

Research Question:

Which Ottawa neighborhoods have the poorest access to public transit, and where should OC Transpo prioritize service expansion?

Data Sources:

1. **OC Transpo Bus Stops** (2024)
 - Source: Ottawa Open Data Portal
 - Format: Regular shapefile
 - Features: bus stops across the City of Ottawa
2. **Ottawa Neighbourhood Boundaries** (Last updated November 2024)
 - Source: Ottawa Neighbourhood Study
 - Format: Polygon shapefile
 - Features: 116 neighborhoods

Software:

- QGIS
- Coordinate Reference System: EPSG:3857 (Web Mercator)

Analysis Workflow:

Step 1: Buffer Creation

- Created 400-meter buffers around all OC Transpo bus stops
- Rationale: 400m represents approximately 5 minutes walking time at average pace (industry standard for transit accessibility)
- Tool: Vector → Geoprocessing → Buffer (with dissolve option enabled)

Step 2: Service Area Calculation

- Dissolved overlapping buffers to create continuous transit service area
- This prevents double-counting areas served by multiple bus stops

Step 3: Coverage Analysis

- Performed intersection between neighborhood boundaries and transit service area
- Calculated percentage of each neighborhood within 400m of transit
- Formula: $(\text{Covered Area} / \text{Total Neighborhood Area}) \times 100$

Step 4: Ranking and Prioritization

- Sorted neighborhoods by coverage percentage (ascending)
- Identified priority tiers:
 - Tier 1: 0% coverage (no access)
 - Tier 2: <50% coverage (critically underserved)
 - Tier 3: 50-75% coverage (poor access)

Assumptions and Limitations:

1. 400m walking distance assumes uniform walkability (may overestimate accessibility in areas without sidewalks)
2. Analysis does not account for service frequency (1 bus/hour $\neq$ 10 buses/hour for actual usability)
3. Population distribution within neighborhoods assumed to be uniform
4. Does not account for temporary service disruptions or construction
5. Focuses on spatial access, not temporal access (time-based routing)